

Chapter 3 Paper 2 QP screenshots + official MS extracts

Paper 2 only. QP pages are direct screenshots from the repo PDF. The repo PDF does not contain Paper 2 mark-scheme pages, so the matching official MS is included as extracted text rather than source screenshots.

P2-1 5054/21/M/J/25 Q5 — longitudinal waves, ripple tank, diffraction

- 5 (a) Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.



Fig. 5.1

- (i) On Fig. 5.1, mark the centre of a compression with the letter C, and mark the centre of a rarefaction with the letter R. [1]
- (ii) Describe the difference between a compression and a rarefaction.
-
-
- [1]
- (b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.
- (i) The wooden bar makes 45 complete oscillations in 1.0 minute.
- Calculate the frequency of the wave produced.

frequency =Hz [1]

- (ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.

State what happens to the speed and what happens to the wavelength of the wave produced.

speed

wavelength

[2]



P2-1 5054/21/M/J/25 Q5 — longitudinal waves, ripple tank, diffraction

(iii) The crests of the water wave move into the shallow region shown in Fig. 5.2.

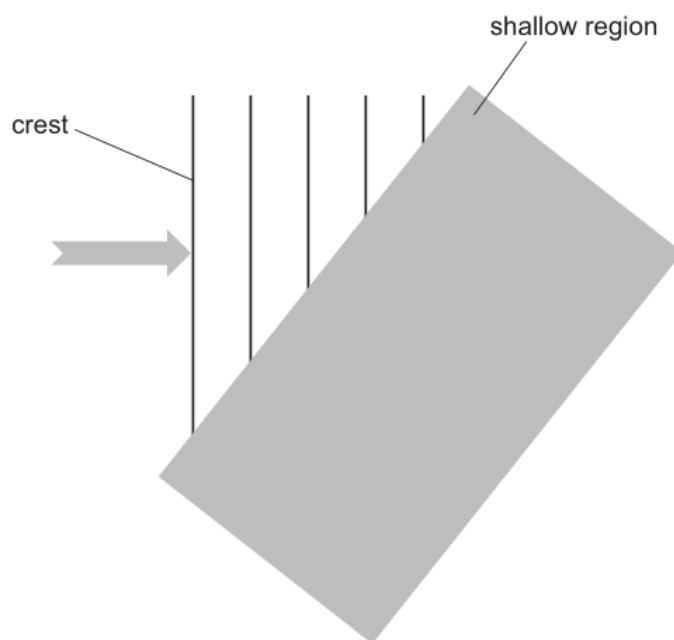


Fig. 5.2

On Fig. 5.2, draw the crests in the shallow region.

[2]

(c) Describe what is meant by the diffraction of a water wave.

.....

.....

..... [1]

[Total: 8]



P2-1 5054/21/M/J/25 Q5 — longitudinal waves, ripple tank, diffraction — mark scheme extract

Official mark scheme extract: 5054_s25_ms_21, Question 5

5(a)(i) C and R marked correctly B1

5(a)(ii) compressions are where pressure is high / particles close together / density is high B1

5(b)(i) 0.75 (Hz) B1

5(b)(ii) speed: no change B1

5(b)(ii) wavelength: decreases B1

5(b)(iii) at least 3 wavefronts in the shallow region parallel to each other and on correct side of normal M1

5(b)(iii) refracted towards the surface on correct side of surface and join wavefronts shown A1

5(c) spreading out / bending at an edge / obstacle OR spreading out after passing through a (small) gap / hole B1

P2-2 5054/21/O/N/14 Q4 — wave motion, frequency, wavelength, ripple-tank reflection

4 Ripple tanks are often used to illustrate wave motion.

(a) Describe what is meant by *wave motion* in a ripple tank.

.....

[2]

(b) When describing wave motion, state what is meant by

(i) *frequency*,

.....[1]

(ii) *wavelength*.

.....
[2]

(c) A water wave in a ripple tank strikes a barrier. Fig. 4.1 shows some wavefronts of the incident wave.

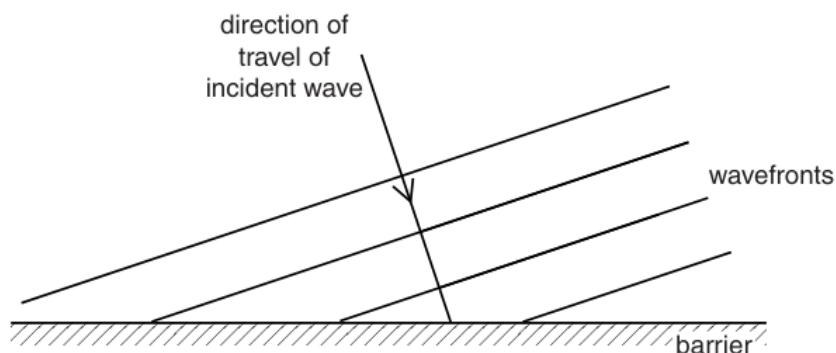


Fig. 4.1

The water wave hits the barrier and is reflected. Three of the wavefronts in Fig. 4.1 have already hit the barrier. The reflected parts of these wavefronts are not shown.

On Fig. 4.1, draw the reflected parts of these three wavefronts.

[3]

P2-2 5054/21/O/N/14 Q4 — wave motion, frequency, wavelength, ripple-tank reflection — mark scheme extract

Official mark scheme extract: 5054_w14_ms_21, Question 4

4(a) any two from: transmission of energy; without net movement of medium; through vibration of particles B2

4(b)(i) number of (complete) waves / cycles / oscillations per unit time / second B1

4(b)(ii) distance between (neighbouring) waves C1

distance between (neighbouring) wavefronts / points of same phase or crest to crest / trough to trough distance A1

4(c) three reflected wavefronts roughly correct direction M1

wavelengths equal to each other and incident wavelength by eye A1

reflected wavefronts joined to incident wavefronts B1

- 5 Optical fibres are used to transmit telephone signals. Fig. 5.1 shows a ray of light travelling along the inside surface of an optical fibre at P.

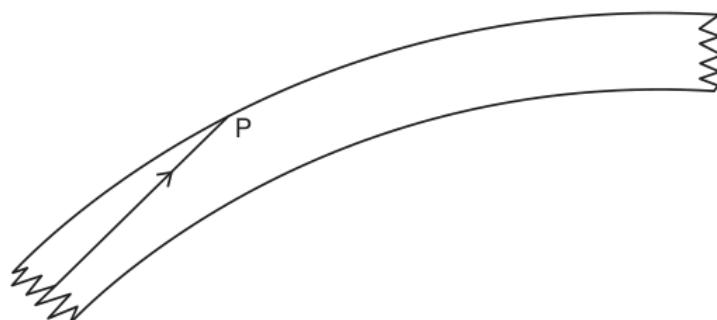


Fig. 5.1

- (a) State one advantage of using optical fibres to transmit telephone signals.

.....
[1]

- (b) (i) On Fig. 5.1, draw a normal at P and mark the angle of incidence with the letter i . [1]

- (ii) State and explain what happens to the ray at P. Use the term *critical angle* in your answer.

.....

[2]

- (c) The optical fibre is made of glass of refractive index 1.5.
 At the start of the optical fibre, the ray enters the glass from air.
 The angle of incidence in the air is 60° .

Calculate the angle of refraction in the glass.

angle =[2]

- 6 Fig. 6.1 shows a wave on the surface of water. The wave is travelling to the right.

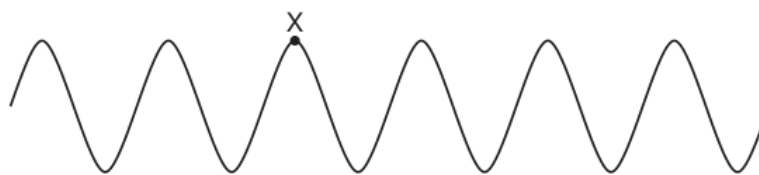


Fig. 6.1 (not to scale)

- (a) Describe what is meant by *wave motion*.

.....

[2]

- (b) On Fig. 6.1, draw an arrow to show the direction of the movement of a water molecule at X. [1]

- (c) The frequency of the water wave is 2.0 Hz and the wavelength is 2.5 cm.

- (i) Calculate the speed of the wave.

speed =[2]

- (ii) On Fig. 6.1, mark a distance which shows how far a wavefront at X moves in 1.0 s. Label this distance *D*. [1]

P2-3 5054/22/M/J/12 Q5-Q6 — optical fibres/TIR + wave motion/ $v=f\lambda$ — mark scheme extract

Official mark scheme extract: 5054_s12_ms_22, Questions 5–6

5(a) more telephone signals at one time OR greater bandwidth/more data per second OR faster transfer OR less attenuation/loss OR more secure OR less interference/higher quality B1

5(b)(i) correct normal and angle marked B1

5(b)(ii) total internal reflection B1; angle of incidence is larger than critical angle B1

5(c) ($n =$) $\sin i / \sin r$ in any form numerical or algebraic C1; $35(.2644)^\circ$ unit $^\circ$ needed A1

6(a) any 2 of: oscillation/vibration up and down; carries energy; no net movement of medium/transfer of matter B2

6(b) arrow downwards or upwards or both B1

6(c)(i) ($v =$) $f\lambda$ in any form; 5.0 cm/s or 0.050 m/s A1

6(c)(ii) line/indication labelled D of length 2 wavelengths B1

P2-4 5054/22/O/N/12 Q10 — law of reflection experiment, optical fibre, plane mirror

- 10 A laser produces red light of frequency 4.7×10^{14} Hz. The speed of light in glass is 2.0×10^8 m/s.

(a) Calculate the wavelength in glass of light from this laser.

wavelength = [2]

(b) Describe an experiment to verify *the law of reflection* for light. You may include a diagram in your answer.

.....

 [5]

(c) Fig. 10.1 shows a ray of light travelling in an optical fibre. The ray strikes the side of the fibre at P.

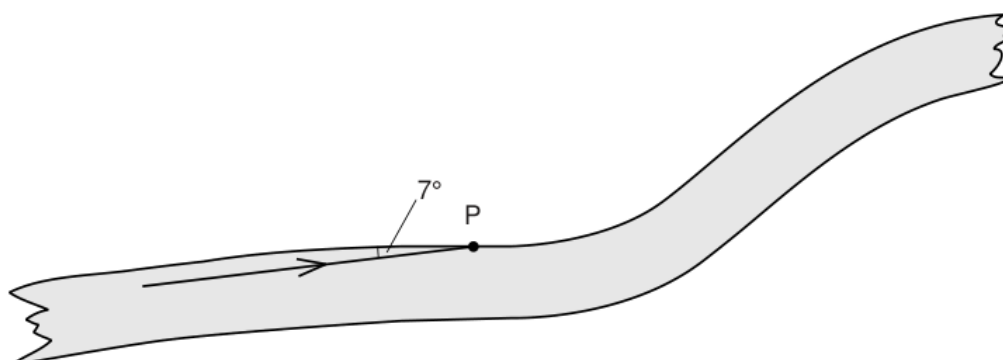


Fig. 10.1

The angle between the ray and the side of the fibre is 7°

P2-4 5054/22/O/N/12 Q10 — law of reflection experiment, optical fibre, plane mirror

- (i) Determine the angle of incidence of the ray at P.

angle =

- (ii) State and explain what happens to the ray at P.

.....

 [2]

- (d) A room is illuminated by wall lamps. Fig. 10.2 shows a mirror on the wall behind one of the lamps.

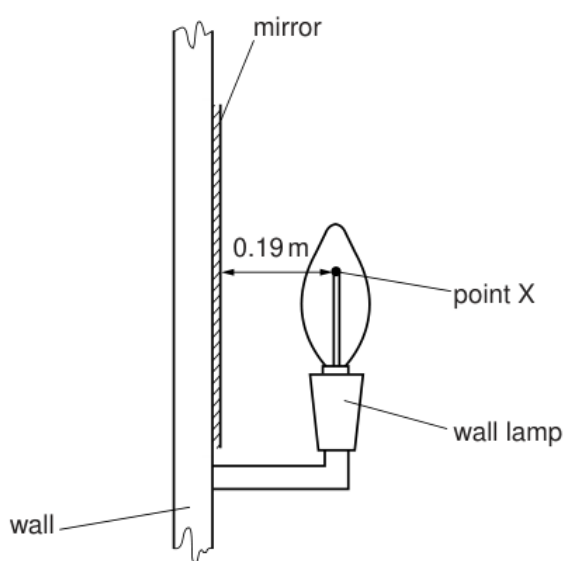


Fig. 10.2 (not to scale)

X is a point on the filament of the lamp. It is 0.19 m in front of the mirror.

- (i) On Fig. 10.2, draw rays from X and locate the image of X. Label the image I. [3]

- (ii) State the distance between I and the mirror.

distance = [1]

- (iii) Suggest one advantage of placing a mirror behind the lamp in the room.

.....
 [1]

P2-4 5054/22/O/N/12 Q10 — law of reflection experiment, optical fibre, plane mirror — mark scheme extract

Official mark scheme extract: 5054_w12_ms_22, Question 10

10(a) ($\lambda =$) v/f or $2 \times 10^8 / 4.7 \times 10^{14}$ C1; 4.3×10^{-7} m A1

10(b) raybox/light source/pins and mirror OR laser and mirror B1; shine ray at mirror/place two pins B1; mark rays/two more pins in line with image B1; measure i and r and equal B1; repeat B1

10(c)(i) 83° B1

10(c)(ii) total internal reflection/TIR B1; angle of incidence exceeds critical angle B1

10(d)(i) at least one ray from X to mirror M1; at least two rays from X to mirror and correct reflections A1; rays traced back to marked I/correct I B1

10(d)(ii) 0.19 m B1

10(d)(iii) less/no light wasted or hall brighter B1

- 10 A student traces the path of a ray of blue light as it enters and as it leaves a glass prism. Fig. 10.1 shows the trace obtained by the student.

For
Examiner's
Use

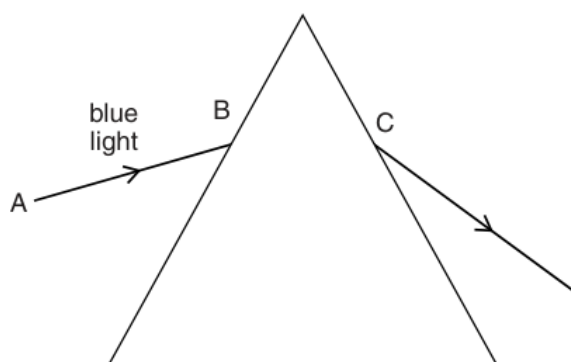


Fig. 10.1

- (a) On Fig. 10.1, draw and label, at the point B, the normal, the angle of incidence i and the angle of refraction r . [3]

- (b) State, in terms of the properties of light waves, why the light refracts at B.

..... [1]

- (c) The angle of incidence for the ray of blue light at B is 45° . The refractive index of the glass is 1.5. Calculate the angle of refraction at B.

angle of refraction = [3]

- (d) The student performs another experiment with a ray of red light along the line AB.

On Fig. 10.1, show the path taken by this ray of light as it passes through and leaves the prism. [2]

- (e) The student performs another experiment with a semicircular glass block and a ray of white light. Fig. 10.2 shows the path taken by this ray of light as it enters the glass at P until it hits the straight edge at Q.

For
Examiner's
Use

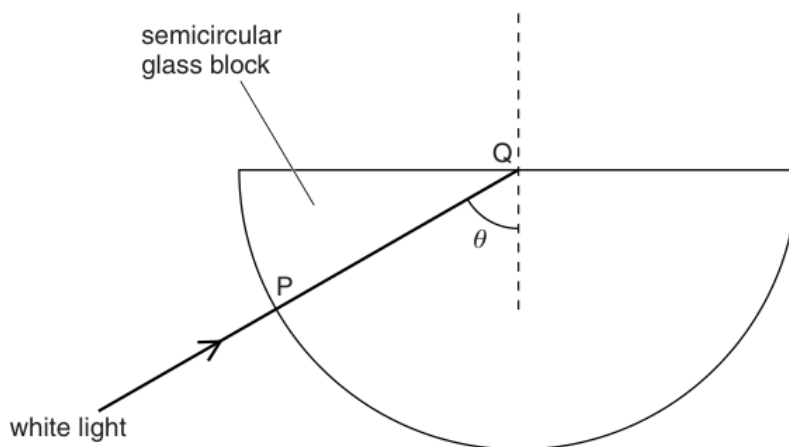


Fig. 10.2

The student finds that there is no change in direction as the ray enters the glass at P and that no light passes out of the glass at Q.

- (i) Explain why the ray does not change direction at P.

.....
 [1]

- (ii) Explain why no light passes out of the glass at Q.

.....

 [2]

- (iii) On Fig. 10.2, draw the complete path followed by this ray. [1]

- (iv) The student directs the ray of white light into the glass along different paths, so that the angle θ is slowly reduced.

Describe what happens to the ray at Q.

.....

 [2]

P2-5 5054/22/M/J/13 Q10 — refraction, Snell's law, critical angle, TIR — mark scheme extract

Official mark scheme extract: 5054_s13_ms_22, Question 10

10(a) correct normal B1; correct angle of incidence B1; correct angle of refraction B1

10(b) decrease/change in speed/wavelength B1

10(c) $n = \sin i / \sin r$ C1; $\sin r = \sin 45^\circ / 1.5$ or 0.4714 seen C1; $28(.1)^\circ$ C1

10(d) refracts less at first face and on correct side of normal B1; refraction at second face away from normal so red and blue diverge B1

10(e)(i) angle of incidence is 0 OR ray along normal/perpendicular to glass B1

10(e)(ii) angle of incidence/ θ is larger than critical angle B1; total internal reflection occurs B1

10(e)(iii) reflected ray correctly and emerging without refraction B1

10(e)(iv) eventually light emerges/refracts out/weak refracted ray appears B1; coloured emergence/correct movement of reflected ray B1

P2-6 5054/21/M/J/23 Q7 — diffraction through gap + wavelength definition

- 7 A water wave in a ripple tank diffracts as it passes through the gap in a barrier.

Fig. 7.1 shows a drawing made by a student of the crests in the pattern.

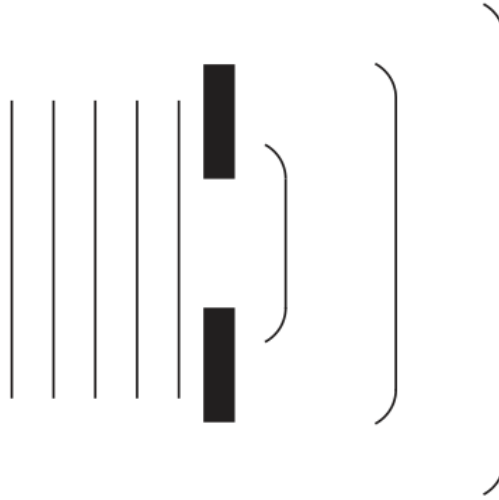


Fig. 7.1

- (a) State **one** way in which the student's drawing of the crests is wrong.

.....
 [1]

- (b) The gap in the barrier is now made smaller than the wavelength, as shown in Fig. 7.2.

Complete Fig. 7.2 with at least three crests to show the new diffraction pattern.

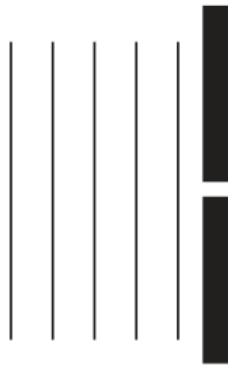


Fig. 7.2

[2]

P2-6 5054/21/M/J/23 Q7 — diffraction through gap + wavelength definition — mark scheme extract

Official mark scheme extract: 5054_s23_ms_21, Question 7

7(a) wavelength has altered/too large OR crests in shadow should be bent/quarter circular above or below gap OR vertical lines too straight on two crests B1

7(b) at least 3 semicircles centered on the gap B1; wavelength constant and same on right/left of gap and some bending B1

7(c)(i) distance between two consecutive identical points such as two consecutive crests B1

7(c)(ii) $v=f\lambda$ numerical/algebraic e.g. $f=6/2$ Hz C1; 3.0 Hz A1

7(c)(iii) 1.3 cm B1

5 Water waves are transverse waves.

(a) Underline **two** other examples of transverse waves.

seismic P-waves

seismic S-waves

sound

X-rays

[1]

(b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

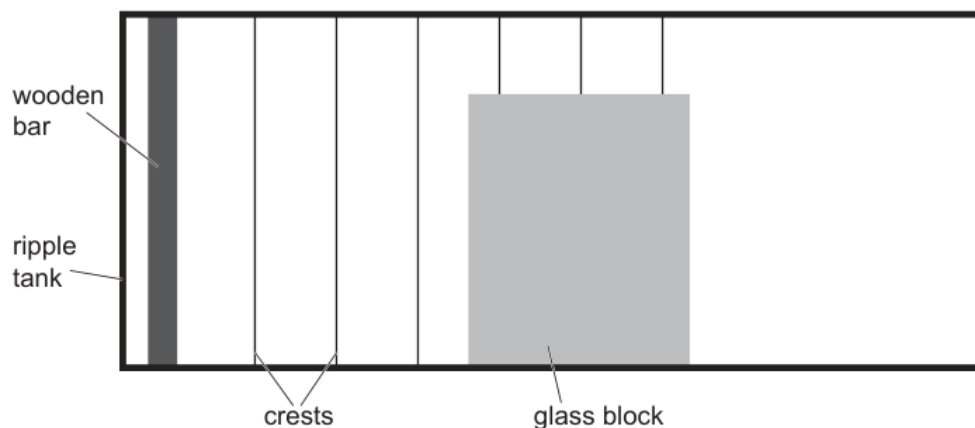


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

(i) The speed of the water wave is 27 cm/s.

Calculate the frequency and the wavelength of the wave.

frequency = Hz

wavelength = cm
[3]

(ii) The wave diffracts at the right-hand edge of the glass block.

On Fig. 5.1 draw **two** crests after they pass the glass block to show the diffraction. [2]

(iii) Describe how a wave with a smaller wavelength is made with the wooden bar.

.....
..... [1]

(iv) Describe how a decrease in wavelength affects the diffraction.

.....
..... [1]

P2-7 5054/22/M/J/23 Q5 — transverse examples and diffraction at an edge — mark scheme extract

Official mark scheme extract: 5054_s23_ms_22, Question 5

5(a) seismic S-waves and X-rays B1

5(b)(i) wavelength = v/f OR distance= vt OR $f=1/T$ or $1/0.15$ C1; frequency 6.7 Hz A1; wavelength 4.0 cm A1

5(b)(ii) straight wavefronts above block to the right with same wavelength B1; correct diffraction wavefronts quarter circles with centre at top corner B1

5(b)(iii) increase frequency of movement of wooden bar B1

5(b)(iv) less diffraction or less bending B1